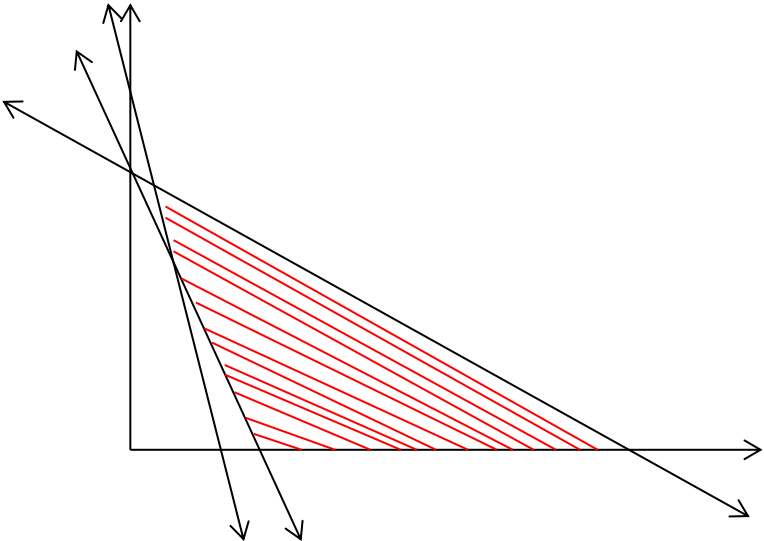


<u>KENDRIYA VIDYALAYA SANGATHAN</u> <u>REGION</u> :- DEHRADUN <u>PRE BOARD</u> -1 <u>SESSION</u> :-2025-26		
<u>SUBJECT:- MATHEMATICS</u>		<u>CODE :- 041</u>
<u>MARKING SCHEME</u>		
SECTION A [1 MARK MULTIPLE CHOICE QUESTIONS]		
Q1	(a) $-\frac{\sqrt{3}}{2}$	1
Q 2	a) 1	1
Q3	(b) $\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$	1
Q4	(b) 4	1
Q5	(c) 13	1
Q6	(c) 64	1
Q7	(b) 2	1
Q8	(a)0	1
Q9	(c) ± 5	1
Q10	(c) (3,3)	1
Q11	(c) 0	1
Q12	(b) $\tan(xe^x) + c$	1
Q13	(a) 2	1
Q14	(d) $\frac{1}{\sqrt{5}}(2\hat{i} + \hat{k})$	1
Q15	(c) $4\sqrt{2}$	1
Q16	(a) p= q	1
Q17	(c) $X+2y \geq 76$, $2x+y \leq 104$, $x,y \geq 0$	1
Q18	(d) $\frac{7}{12}$	1

Q19	(c) A is true but R is false.	1
Q20	(a) Both (A) and (R) are true and (R) is the correct explanation of (A)	1
SECTION B This section comprises of 5 very short answer (VSA) type questions of 2 marks each.		
Q 21	$\begin{aligned} & \tan^2(\sec^{-1}2) + \cot^2(\operatorname{cosec}^{-1}3) \\ &= \sec^2(\sec^{-1}2) - 1 + \operatorname{cosec}^2(\cos 3) - 1 \\ &= (2)^2 - 1 + (3)^2 - 1 \\ &= 4 - 1 + 9 - 1 = 11. \end{aligned}$ Or $\begin{aligned} & \cot^{-1} \left(\frac{\sqrt{1 + \sin x} + \sqrt{1 - \sin x}}{\sqrt{1 + \sin x} - \sqrt{1 - \sin x}} \right) \\ &= \cot^{-1} \left(\frac{\sqrt{\left[\cos \frac{x}{2} + \sin \frac{x}{2}\right]^2} + \sqrt{\left[\cos \frac{x}{2} - \sin \frac{x}{2}\right]^2}}{\sqrt{\left[\cos \frac{x}{2} + \sin \frac{x}{2}\right]^2} - \sqrt{\left[\cos \frac{x}{2} - \sin \frac{x}{2}\right]^2}} \right) \\ &= \cot^{-1} \left(\frac{\left[\cos \frac{x}{2} + \sin \frac{x}{2}\right] + \left[\cos \frac{x}{2} - \sin \frac{x}{2}\right]}{\left[\cos \frac{x}{2} + \sin \frac{x}{2}\right] - \left[\cos \frac{x}{2} - \sin \frac{x}{2}\right]} \right) \\ &= \cot^{-1} \left(\frac{2 \cos \frac{x}{2}}{2 \sin \frac{x}{2}} \right) = \cot^{-1} \left(\cot \frac{x}{2} \right) = \frac{x}{2} \end{aligned}$	1 1 1 1
Q22	$\log\left(\frac{\sin x}{1 + \cos x}\right) = \log\left(\frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}}\right) = \log\left(\tan \frac{x}{2}\right)$ $\frac{d}{dx} \left(\log\left(\tan \frac{x}{2}\right) \right) = \frac{1}{\tan \frac{x}{2}} \sec^2 \frac{x}{2} \cdot \frac{1}{2} = \frac{2}{2 \sin \frac{x}{2} \cos \frac{x}{2}} = \frac{2}{\sin x} = 2 \operatorname{cosec} x$	1 1
Q23	$\begin{aligned} & 3x^2 + \sin 6x = t \\ & \text{Differentiate w.r.t } x \\ & (6x + 6 \cos 6x) = \frac{dt}{dx} \\ & (6x + 6 \cos 6x) dx = dt \\ & \frac{1}{6} \log 3x^2 + \sin 6x \end{aligned}$ or $\begin{aligned} & \int_0^\pi \cos x dx \\ &= \int_0^{\frac{\pi}{2}} \cos x dx - \int_{\frac{\pi}{2}}^\pi \cos x dx \\ &= [-\sin x]_0^{\frac{\pi}{2}} + [\sin x]_{\frac{\pi}{2}}^\pi \\ &= -2 \end{aligned}$	1 1 1 1

Q24	Apply L.H.L = R.H.L Get $a+b = 3$ L.H.D = $2a$ R.H.D = 2 Simplify and get $a = 1$ and $b=2$	1 1
Q25	Each vector is perpendicular to other so getting result $[\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}] = 0$ $\vec{a} + \vec{b} + \vec{c} ^2 = \vec{a} ^2 + \vec{b} ^2 + \vec{c} ^2 + 2[\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}]$ $\vec{a} + \vec{b} + \vec{c} = 5\sqrt{2}$	1 1
SECTION C This section comprises of 6 short answer (SA) type questions of 3 marks each.		
Q26	$\frac{dy}{d\theta} = a \cos \theta$ $\frac{dx}{d\theta} = a(\operatorname{cosec} \theta - \sin \theta)$ $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\cos \theta}{(\operatorname{cosec} \theta - \sin \theta)} = \tan \theta$ $\left(\frac{dy}{dx}\right) \text{ at } \theta = \frac{\pi}{4} = 1$ OR $Y = x^x$ $\log y = x \log x$ Differentiate both sides with respect to x $\frac{dy}{y dx} = [1 + \log x]$ $\frac{dy}{dx} = y[1 + \log x]$ Again differentiate with respect to x Simplify and getting $\frac{d^2 y}{dx^2} - \frac{1}{y} \left(\frac{dy}{dx}\right)^2 - \frac{y}{x} = 0.$	1 1 1 1 1 1
Q27:-	$\frac{dr}{dt} = -3 \text{ cm/min}$ $\frac{dh}{dt} = 2 \text{ cm/min}$ $V = \pi r^2 h$ $r = 7 \text{ cm}, h = 2 \text{ cm}$ $\frac{dv}{dt} = \pi \left[r^2 \frac{dh}{dt} + 2rh \frac{dr}{dt} \right]$ $= \pi [98 - 84]$ $= 14 \pi = 44 \text{ cm}^3/\text{min}$	1 1 1

Q28:-	Figure Area of bounded region = $2 \int_0^4 \sqrt{y} dx = 2x \frac{2}{3} [y^{3/2}]_0^4 =$ $\frac{4}{3} \times 8 = \frac{32}{3}$ sq units OR Figure $A = \int_2^8 y dx$ $A = \int_2^8 \frac{-x+8}{2} dx$ Solve Getting answer 9 sq units	1 1 1 1 1 1
Q29:-	To find direction ratios of the required line with different methods Direction ratios are 2, -7, 4 To find the equation of required line $\frac{x+1}{2} = \frac{y-3}{-7} = \frac{z+2}{4}$ OR To find $\lambda = -1$ foot of perpendicular $\hat{i} + 2\hat{j} + 3\hat{k}$ Length of foot of perpendicular = $\sqrt{14}$	$1\frac{1}{2}$ $1\frac{1}{2}$ 2 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
Q30:-	 Corner points of feasible region are (15,0), $Z = 20 \times 15 + 10 \times 0 = 300$ (40,0), $Z = 20 \times 40 + 10 \times 0 = 800$ (4,18), $Z = 20 \times 4 + 10 \times 18 = 260$ (6,12), $Z = 20 \times 6 + 10 \times 12 = 240$ Minimum value of Z is 240 at point (6,12)	2 1

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Q33	$I = \int \sqrt{\frac{\sin(x-\alpha) \times \sin(x-\alpha)}{\sin(x+\alpha) \times \sin(x-\alpha)}} dx$ $I = \int \frac{\sin(x-\alpha)}{\sqrt{\sin^2 x - \sin^2 \alpha}} dx$ $I = \int \frac{\sin x \cos \alpha - \cos x \sin \alpha}{\sqrt{\sin^2 x - \sin^2 \alpha}}$ $I = \cos \alpha \int \frac{\sin x}{\sqrt{1 - \cos^2 x - 1 + \cos^2 \alpha}} - \sin \alpha \int \frac{\cos x}{\sqrt{\sin^2 x - \sin^2 \alpha}}$ Solve 1st integral and 2nd integral and getting final answer $I = -\cos \alpha \sin^{-1} \left(\frac{\cos x}{\cos \alpha} \right) - \sin \alpha \log \sin x + \sqrt{\sin^2 x - \sin^2 \alpha} + C$ OR $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x + \cos x}{\sqrt{1 - (\cos x - \sin x)^2}} dx$ Take $\cos x - \sin x = t$ and $-(\sin x + \cos x) dx = dt$ Lower limit for $t = \frac{\sqrt{3}-1}{2}$ Upper limit for $t = \frac{1-\sqrt{3}}{2}$ $I = -\int_{\frac{\sqrt{3}-1}{2}}^{\frac{1-\sqrt{3}}{2}} \frac{1}{\sqrt{1-t^2}} dt = 2 \sin^{-1} \left(\frac{\sqrt{3}-1}{2} \right)$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 1+1 1 OR 1 1 1+1+1
Q34:-	$(x^2 + xy) dy = (x^2 + y^2) dx$ $\frac{dy}{dx} = \frac{(x^2 + y^2)}{(x^2 + xy)}$ Putting $y = vx$ and $\frac{dy}{dx} = v + x \frac{dv}{dx}$ We get $\frac{dx}{x} = \left(\frac{2}{1-v} - 1 \right) dv$ Integration both sides And getting answer $\log x + \log C + 2 \log 1-v = -v$ after substitution $x=1$ and $y=0$ value of $C=1$ final particular solution $(x-y)^2 = x e^{-y/x}$ OR $x \frac{dy}{dx} + y - x + xy \cot x = 0$ $\frac{dy}{dx} + \left(\frac{1}{x} + \cot x \right) y = 1$ Compare it with basic form $\frac{dy}{dx} + Py = Q$ $P = \left(\frac{1}{x} + \cot x \right), Q = 1$ I.F = $x \cdot \sin x$ Integration Final answer $Y \times \sin x = -x \cos x + \sin x + c$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ 2 1 1/2 OR 1 $\frac{1}{2}$ 1 2+1/2

Q35:-	$\vec{r} = (4\hat{i} - \hat{j}) + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$ and $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(\hat{2i} + 4\hat{j} - 5\hat{k})$ $a_1 = (4\hat{i} - \hat{j}), b_1 = (\hat{i} + 2\hat{j} - 3\hat{k}) \quad a_2 = (\hat{i} - \hat{j} + 2\hat{k}), b_2 = (\hat{2i} + 4\hat{j} - 5\hat{k})$ $d = \frac{ (\vec{a_2} - \vec{a_1}) \cdot (\vec{b_2} \times \vec{b_1}) }{ \vec{b_2} \times \vec{b_1} }$ $(\vec{a_2} - \vec{a_1}) = (-3\hat{i} + 0\hat{j} + 2\hat{k})$ $(\vec{b_2} \times \vec{b_1}) = (2\hat{i} - \hat{j} + 0\hat{k})$ $(\vec{a_2} - \vec{a_1}) \cdot (\vec{b_2} \times \vec{b_1}) = -6$ $ \vec{b_2} \times \vec{b_1} = \sqrt{5}$ $S.D = \frac{6}{\sqrt{5}}$	$\frac{1}{2}$ 1 $\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$ 1
<u>SECTION E</u>		
Q 36	i) Since every number is divisible by itself, So (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6) . So, R is reflexive relation on B. ii) X, y, z are three numbers in set B $(x, y) \in R \Rightarrow y$ is divisible by $x \Rightarrow y = kx \rightarrow (1)$ $(y, z) \in R \Rightarrow z$ is divisible by $y \Rightarrow z = py \rightarrow (2)$ From (1) and (2) $Z = pkx \Rightarrow z$ is divisible by $x \Rightarrow (x, z) \in R$ $(x, y), (y, z) \in R \Rightarrow (x, z) \in R$ so this relation is transitive iii) Symmetricity :- $(2, 4) \in R$ but $(4, 2) \notin R \Rightarrow R$ is not symmetric Relation R is reflexive and transitive but not symmetric so this relation is not equivalence relation OR $R = \{(1, 2), (2, 2), (1, 3), (3, 4), (3, 1), (4, 3), (5, 5),\}$ Since $(1, 1) \notin R$, so R is not reflexive. Hence 'R' is not an equivalence relation.	1 1 2 2

Q37:-	i) $V = x \cdot x \cdot y = x^2y$ ii) $A = x^2 + 4xy$ iii) $y = \frac{A-x^2}{4x}$ substitute in equation 1 $v = \frac{1}{4} (Ax - x^3)$ $\frac{dv}{dx} = \frac{1}{4} (A - 3x^2)$ at extreme point $\frac{dv}{dx} = 0$ $A - 3x^2 = 0 \Rightarrow x = \sqrt{\frac{A}{3}}$ $\frac{d^2v}{dx^2} = \frac{1}{4}(-6x) < 0$ volume will be maximum when $x = \sqrt{\frac{A}{3}}$ OR $V = x^2y$ $x^2y = 1024 \rightarrow (1)$ cost function $z = 5x^2 + 10xy$ z in terms of x $Z = 5x^2 + \frac{10240}{x}$ $\frac{dz}{dx} = 10x - \frac{10240}{x^2}$ At extreme points $\frac{dz}{dx} = 0$ Getting $x = 8(2)^{1/3}$	1 1 2 2
Q38:-	(a) 0.9 (b) 0.01 (c) 0.999 (d) 0.083	1 1 1 1